Diagnostic addendum: complete saved diagnostics and analytic qualifications

This addendum corrects an evidence-presentation claim in the compact Supplement. Its printed paired-loss table contains summary columns rather than every saved diagnostic field. Its adverse table lists universal-simplification failures rather than every exception to the primary loss-sign screens. The complete saved fields and primary-screen exceptions are displayed here without refitting or changing any result. The accepted compact Supplement and preference-frontier technical document remain byte-for-byte preserved under their original checks. The following analytic qualifications narrow the interpretation of history patterns, horizon coherence and the financial information convention; they add no empirical computation.

Current evidence routes are given by the current extended-results index and current reproduction guide. The older generic branch links in the protected Supplement describe their historical line. The original formal drivers require local design objects, and the simulation driver also enforces its expired original window. Document compilation and saved-field checks do not remove these guards or constitute independent financial replication.

D1 Complete primary-screen exceptions and saved paired-loss diagnostics

These displays add no analysis. They expose existing exceptions and diagnostic fields that were summarized in the compact Supplement. Original source values, decisions and archives remain unchanged.

Every exception to the two primary simulation screens

Table D1. Complete saved primary-screen exceptions.

Criterion	ϕ	W	H	Model	Δ (%)	MCSE
P1	0.7	720	24	QR33	−1.34646	8.29105
P2	0.7	720	6	QR4	+7.31298	4.50282
P2	0.9	720	6	QR4	+0.898697	3.25074
P2	0.9	2880	6	QR33	+0.0346237	3.09686

All listed cases use Student innovations. A nonpositive loss change contradicts the short-history strict-loss prediction; a nonnegative change contradicts the long-history strict-win prediction. The small positive long-history exception remains a strict-win failure, not a rounded tie. MCSE is the saved repetition-cluster uncertainty summary, not a new significance test. Pooled screen decisions remain unchanged. The compact Supplement separately retains every failed universal-simplification inequality. Complete candidate and repetition records remain in the public simulation archive with its member index.

Every saved financial paired-loss diagnostic

Table D2. Full saved HAC fields for the financial learning curve.

H	W	Model	L	Mean difference	LRV	DM t	Nominal p
6	360	QR33	10	13893.2	9.40901×10^{11}	1.1658	0.243695
6	360	QR4	10	−3995.71	3.51487×10^{11}	−0.54857	0.583301
6	720	QR33	10	−9368.96	1.53305×10^{12}	−0.615893	0.537965
6	720	QR4	10	6535.09	9.8903×10^{11}	0.53486	0.592747
6	1440	QR33	10	−28927.4	2.50098×10^{11}	−4.70813	2.50002×10^{-6}
6	1440	QR4	10	−14585.4	9.80038×10^{10}	−3.79219	0.000149325
6	2880	QR33	10	−66848.1	8.60453×10^{11}	−5.86568	4.47296×10^{-9}
6	2880	QR4	10	−19417.7	3.77009×10^{11}	−2.57404	0.010052
6	5760	QR33	10	−28936.6	1.4079×10^{11}	−6.27704	3.45086×10^{-10}
6	5760	QR4	10	−21976.3	5.73228×10^{11}	−2.36256	0.018149
24	360	QR33	24	198276	2.41117×10^{13}	3.28662	0.00101397
24	360	QR4	24	135110	1.97771×10^{13}	2.47285	0.0134039
24	720	QR33	24	248067	3.92526×10^{13}	3.22276	0.00126963
24	720	QR4	24	130027	2.31015×10^{13}	2.20194	0.0276694
24	1440	QR33	24	69647.2	1.98173×10^{13}	1.27343	0.202866
24	1440	QR4	24	−19771.8	3.57064×10^{12}	−0.851662	0.394402
24	2880	QR33	24	−28686.1	2.14543×10^{13}	−0.504089	0.614199
24	2880	QR4	24	−22089.7	1.29457×10^{13}	−0.499713	0.617277
24	5760	QR33	24	14096.7	2.35761×10^{13}	0.236306	0.813195
24	5760	QR4	24	1201.99	1.6773×10^{13}	0.0238886	0.980941

Every row uses 6625 common evaluation issues. The difference is regression minus matched historical headroom, in native FX loss units; L is the saved Bartlett bandwidth and LRV its saved long-run variance. Fields are copied, not recomputed. The tests use a two-sided normal reference, remain post-result and nominal, and carry no family or full-selection adjustment. 0^u, if present, denotes a saved floating-point zero rather than an exact probability. Machine-readable complete contrasts preserve additional scores, breaches and exact saved precision.

D2 Analytic qualifications for history coordinates and horizon coherence

This note qualifies the interpretation of the existing comparisons and corrects an overly broad horizon-repair statement in the earlier Supplement. The arguments are algebraic or distributional: they introduce no fitted forecasts, new empirical scores, resampling output or changes to the archived results.

A. Method-specific cutpoints cannot distinguish constant rescalings

Let W count completed fitting-label origins, let H be the forecast horizon, and let $p_m > 0$ be the declared coefficient count for a fixed method m. Consider the two candidate coordinates

$$c = \frac{W}{H+1}, \qquad b = \frac{W}{(H+1)p_m} = \frac{c}{p_m}.$$

The pooled discovery classifier can distinguish these coordinates when methods have different p_m. The later use of a separate cutpoint for each method has a different invariance. Within a method, positive constant scaling preserves the ordering and equality groups of the coordinate values. If a zero crossing between adjacent points (x_0, y_0) and (x_1, y_1) is interpolated as

$$r = x_0 + (x_1 - x_0)\frac{y_0}{y_0 - y_1},$$

replacing every x by x/p_m replaces r by r/p_m. Crossing directions, crossing counts and censoring statuses are unchanged. Applied to each existing resampling draw separately, the same identity scales the cutpoint-interval endpoints; zero and unbounded endpoints retain their statuses.

Writing $r_b = r_c/p_m$, the subsequent method-specific screen regions obey

$$b < r_b/2 \iff c < r_c/2, \qquad b > 2r_b \iff c > 2r_c.$$

Their interval transfers back to history are also identical:

$$(H+1)p_m[\ell_b, u_b] = (H+1)[\ell_c, u_c].$$

Consequently these later screens and interval transfers cannot select between the two normalizations when each method receives its own cutpoint. This is an exact-arithmetic property of the definitions, rather than a claim about an alternate floating-point execution. It does not undo the specified screen outcomes, establish a complexity mechanism, or resolve the separate comparison between larger and smaller forecasting procedures.

B. A pooled discovery root is not a fixed-horizon learning root

The discovery construction pools a method's simulation cells across horizons and process settings before grouping and sorting by its selected coordinate. A crossing interpolated between neighboring points that differ in both W and H is therefore a cutpoint of that pooled grid summary. By itself it does not identify the amount of history at which a fixed-horizon learning curve crosses zero. Correct execution of the interpolation rule does not remove this distinction.

For a logical example, suppose relative loss is a function of W alone, positive at shorter histories and negative at the longest history. Sorting the observations by $W/(H+1)$ can still place opposite-sign observations from different horizons next to each other and produce an interpolated crossing. Such a crossing does not isolate an effect of the horizon normalization. This is an

identification counterexample, not an estimated alternative process. A grid without common coordinate values across its horizons also supplies no direct comparison of distinct horizons at the same coordinate.

The financial learning curves instead hold a horizon and evaluation record fixed while changing W, together with the corresponding historical reference. Agreement with an interval transferred from the pooled discovery root is a conditional consistency check between these different objects. It does not turn the pooled root into a horizon-specific discovery root. For background, Viering and Loog, Section 2.1 distinguish a fixed learning problem from averages across problems; that distinction does not itself validate the present coordinate.

C. Matched completed-label pools do not match every information input

The matched procedures share the specified rolling pool of completed fitting labels and the refit clock. Conditional features additionally consult issue-known raw lookback observations, while the regression's penalty selection uses its specified validation segment. Thus W counts mature response origins rather than every raw observation consulted for predictors or every observation used for tuning.

The results compare forecasting procedures matched on those declared dimensions. Equal W alone does not isolate architecture from all other information and tuning choices. This qualification does not assert future-label leakage or replace any result. The broader treatment of estimation and data use as components of a forecasting method is consistent with Giacomini and White (2006); the present information-footprint distinction concerns the specified procedures in this study.

D. Sorting crossing horizon reports weakly improves aggregate pinball loss

At one issue origin, let the realized short- and long-horizon peaks satisfy $y_s \le y_l$. Suppose the already available reports cross: the short-horizon report is a, the long-horizon report is b, and $a > b$. Use the same quantile level $0 < \tau < 1$, common units and the equally weighted sum of the two pinball losses. Sorting the reports into horizon order replaces (a, b) by (b, a).

The identity $\rho_\tau(y - c) = \tau(y - c) + (c - y)_+$ gives the original sum minus the sorted sum as

$$\begin{aligned} D &= \rho_\tau(y_s - a) + \rho_\tau(y_l - b) - \rho_\tau(y_s - b) - \rho_\tau(y_l - a) \\ &= (a - y_s)_+ - (b - y_s)_+ - (a - y_l)_+ + (b - y_l)_+ \\ &= \int_b^a \big[\mathbf{1}\{u > y_s\} - \mathbf{1}\{u > y_l\}\big]\, du \\ &= \text{length}\big([b, a] \cap [y_s, y_l]\big) \ge 0. \end{aligned}$$

The inequality is strict exactly when the intersection has positive length. The transformation uses only the issued reports; future outcomes enter the proof of its score property, not the transformation. If both reports satisfy the common observed floor B, their sorted versions still satisfy it. The same statement holds after averaging the sum over common issue origins.

Correction to the earlier Supplement. Its statement that raising a long-horizon report can worsen that report's pinball loss is correct. Its following assertion that target ordering does not identify a universally beneficial repair is too broad when read as an assertion about aggregate equal-weight pinball loss: the identity above provides such a weak improvement. The earlier counterexample rules out a raise-only marginal-loss guarantee, rather than this sorting result. The protected Supplement and its crossing counts are retained; this note explicitly supplies the qualification.

This aggregate result provides no guarantee for each horizon's loss separately, arbitrary unequal weights, differently normalized horizon scores, breach counts or conditional coverage. No sorted empirical forecasts or new financial loss summaries are reported here. Related rearrangement work gives useful context: Chernozhukov, Fernández-Val and Galichon, Proposition 4 establishes an L^p estimation-error improvement for a quantile curve rearranged over its probability index. That proposition is not the fixed-τ, cross-horizon, samplewise pinball identity proved here. No new general rearrangement theorem is claimed.

E. Horizon ordering uses a common lower-inverse selection

For nested peaks under the same issue-information law, $Y_{t,H_s} \le Y_{t,H_l}$ implies $F_s(z \mid \mathcal{F}_t) \ge F_l(z \mid \mathcal{F}_t)$ for every z. At the same level τ, using the common lower-inverse convention

$$q_\tau^-(H \mid \mathcal{F}_t) = \inf\{z : F_H(z \mid \mathcal{F}_t) \ge \tau\}$$

therefore yields $q_\tau^-(H_s \mid \mathcal{F}_t) \le q_\tau^-(H_l \mid \mathcal{F}_t)$. Atoms and flat CDF intervals are allowed. This is the canonical selection already used in the Supplement.

The full quantile functional can instead be set-valued. For example, if the two peaks coincide with a variable taking zero with probability τ and one with probability $1-\tau$, every report in $[0, 1]$ minimizes expected pinball loss. Choosing different members of this interval at different horizons can produce crossing optimal reports. Consequently crossing is a coherence diagnostic relative to a declared selection convention; by itself it does not establish marginal misspecification. Ehm et al., Sections 2.1–2.2 define the potentially set-valued quantile functional and its consistent scoring functions. This selection qualification is distinct from both the aggregate sorting identity and the strict-tail coverage inequality for a true conditional lower-inverse quantile.

F. Observation and calibration contracts

In the weekly experiment, the current-position floor is observed under a stipulated economic-date replay of one downloaded vintage. This convention makes the current and prior snapshot values available at each issue index. It is distinct from the historical information of a public observer: the New York Fed source page describes publication in the following week, and later versions may revise earlier observations. Delaying only the predictors would not repair a public-information interpretation while retaining an unreleased current-position floor. No release-vintage reconstruction or changed financial experiment is supplied by this clarification.

The main delayed-calibration comparison concerns the score advantage of the wrapped regression over the wrapped historical reference. Each wrapper's saved mean loss is higher than that of its own unwrapped base. Thus a retained relative advantage and closer observed breach frequencies coexist with a score cost relative to the bases; this is not a claim of joint absolute improvement. The validation warmup and calibration-parameter choice are the already documented retrospective second-stage selection, not a fresh historical issue-by-issue model-selection exercise.

The earlier Supplement's sentence about state carrying across the unissued gap refers to the selected ACI candidate's calibration state. It should not be read as a claim that the headroom-SAV fitted state is carried from validation into evaluation. That recursion is refitted on its evaluation segment's specified completed labels and refit grid. These distinctions follow the frozen implementations and existing saved outputs; no canonical ACI theorem, CAViaR-family result, new score or statistical comparison is imported.